R-Series RTD

3-Wire Platinum 100 ohm at 0° C

DESCRIPTION

R-Series RTDs are a family of 3-wire Platinum 100 Ohm at 0° C RTDs used in pairs as inputs for the Badger Meter Impeller line of BTU monitors. BTU monitors require two temperature sensors and a flow sensor to operate. When operating BTU monitors, temperature sensor selection and placement is extremely important. There are a number of options available to help maximize the efficiency of BTU monitoring. R-Series RTDs are designed for use with liquid heating or cooling media.

An RTD is a special kind of resistor made of specific semiconductor material that increases resistance as temperature increases.

ACCURACY AND TEMPERATURE

Accuracy is ±0.3 + 0.005*(t)° C over a range of –100…149° C (–148…300° F), with maximum temperature of 260° C (500° F). Stability is better than 0.2° C after 10,000 hours.

The following chart shows the relationship between resistance and temperature.

°C	°F	Resistance
0	32	100.00 Ω
10	50	103.90 Ω
20	68	107.79 Ω
30	86	111.67 Ω
40	104	115.54 Ω
50	122	119.40 Ω
60	140	123.24 Ω
70	158	127.08 Ω
80	176	130.90 Ω
90	194	134.71 Ω
100	212	138.51 Ω

DIFFERENTIATOR

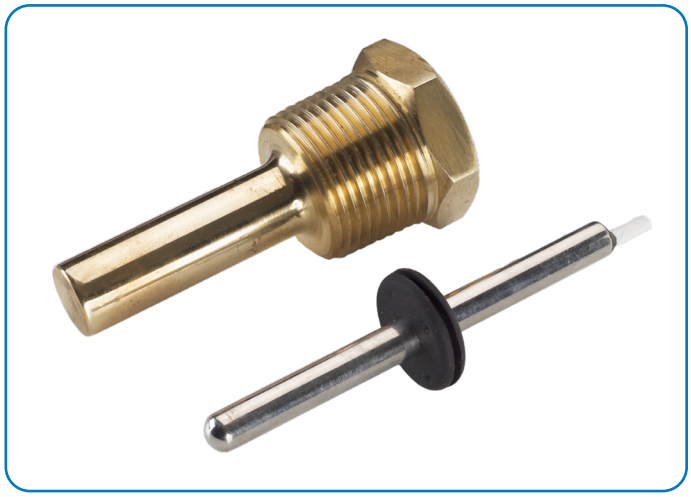
Because of their broader accuracy range, RTDs are a better choice than thermistors when the operating temperatures are going to be below 0° C (32° F) or above 70° C (158° F).

CONSTRUCTION

The temperature probe itself is a 1/4 in. (6.35 mm) diameter 3-1/2 in. (88.90 mm) long stainless steel tube, which is potted solid.

R-Series RTDs use thin film technology and are precision cut to 100 Ohm resistance at 0° C using material that complies with the IEC 751 Tolerance class B Standard.

R-Series RTDs are not sold as matched pairs; however, since they utilize thin film technology that is laser trimmed during production to 100.00 Ohm ±1% at 0° C with a high purity substrate, they are interchangeable. As a result, sensors can be individually replaced without affecting the accuracy of the BTU calculations, with no circuitry adjustments. However, since all RTDs drift over time, it is always a good idea to replace them in pairs.



The maximum distance between a temperature sensor and a BTU monitor or transmitter is 500 feet. Since these RTDs are 3-wire devices, the lead compensating circuitry in Badger Meter Impeller monitors and transmitters compensates for the lead resistance in one current carrying wire, and assumes the other current carrying lead is the same length.

APPLICATIONS

The selection and placement of the temperature sensors is extremely important to the operation of a BTU monitor. For smaller size pipes, they are sold as direct insert with compression fittings. For larger pipe sizes Thermowell versions are available.

Direct Insert Configuration

For smaller pipe sizes, the R-Series is available with a compression fitting.

These have the advantage of lower cost and variable insertion depth, but require service interruption.

Thermowell Style

Use thermowells for larger pipe sizes or pressure systems.

A thermowell is a sleeve assembly that protects the temperature sensing device from direct damage, and allows replacement of the RTD probe itself without draining fluid or interfering with system operation.

Ideally, the sensing area, or tip, of the thermowell should be as close to the actual center of the pipe I.D. as practical. This helps the response time, and makes the reading more representative of the actual fluid temperature. Badger Meter Impeller offers a number of optional thermowells. For the longer versions, the T-Series includes spacer/spring assemblies to make sure the RTD element is held at the bottom of the thermowell, for maximum heat transfer. Usually, some type of heat transfer material (silicone grease) is used to fill the area between the I.D. of the thermowell and the O.D. of the thermistor tubes, again to maximize heat transfer.

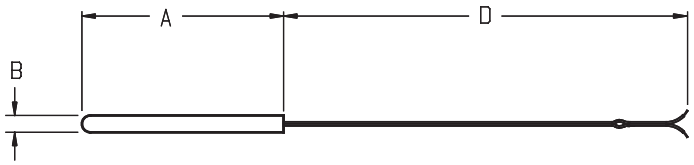
XMT-DS-02807-EN-02 (December 2021)

Product Data Sheet

DIMENSIONS

Temperature Probe

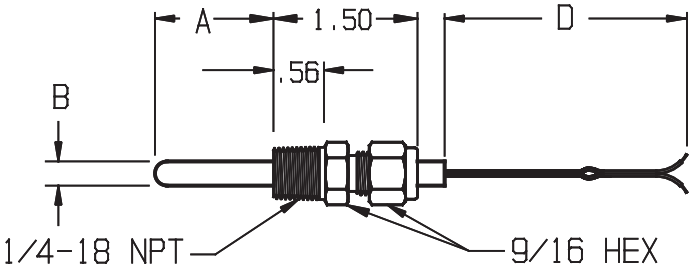
R-100



Model	Material	A	B	D
R-100	Stainless Steel	3.5 in.	0.25 in.	12 in.

Direct Insert Configuration

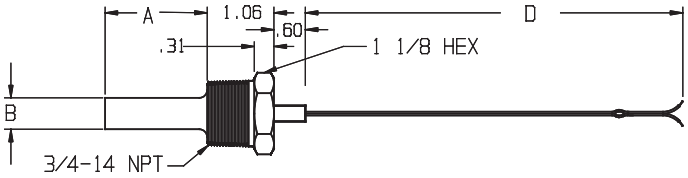
R-106B and R-106S



Model	Material	A	B	D
R-106B	Brass	0…1.25 in.	0.25 in.	12 in.
R-106S	Stainless Steel	0…1.25 in.	0.25 in.	12 in.

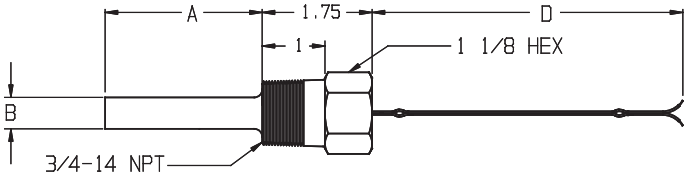
Thermowell Style

R-116B and R-116S



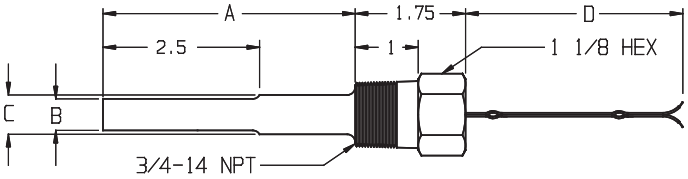
Model	Material	A	B	D
R-116B	Brass	1.625 in.	0.50 in.	12 in.
R-116S	Stainless Steel	1.625 in.	0.50 in.	12 in.

R-125



Model	Material	A	B	D
R-125	Stainless Steel	2.50 in.	0.50 in.	10 in.

R-140, R-160 and R-180



Model	Material	A	B	C	D
R-140	Stainless Steel	4.0 in.	0.50 in.	0.750 in.	12 in.
R-160	Stainless Steel	6.0 in.	0.50 in.	0.750 in.	12 in.
R-180	Stainless Steel	6.0 in.	0.50 in.	0.750 in.	12 in.

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